Capital and Labor Shares in Healthcare

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February 2026

Introduction

Motivation

- Healthcare is a large and growing share of the U.S. economy
- Understanding the production technology requires measuring factor shares
- How much of healthcare value added goes to labor vs. capital?
- Key for: cost growth decomposition, technology adoption, policy analysis

Measuring Factor Shares in Healthcare

Roadmap

1. **From production to value added** — the firm's problem, factor payments, and value added
2. **National accounts framework** — mapping theory to BEA data
3. **Raw factor shares** — labor, capital, and tax shares by industry
4. **The mixed income problem** — proprietors' income contains both labor and capital
5. **Gollin (2002) adjustments** — three methods for allocating mixed income
6. **Sensitivity analysis** — nonprofit handling, gross vs. net of depreciation
7. **Key results** — healthcare vs. other industries

The Firm's Production Function

A firm in industry i produces output Y_i using capital K_i and labor L_i:

$$Y_i = A_i \, F(K_i, \, L_i)$$

where A_i is total factor productivity (technology, management, regulation).

In Healthcare

- **Labor** (L): physicians, nurses, technicians, administrative staff
- **Capital** (K): hospital buildings, MRI machines, IT systems, medical devices

A common parametric form is **Cobb-Douglas**:

$$Y_i = A_i \, K_i^{\alpha} \, L_i^{1-\alpha}$$

where α is the capital share parameter — the object we want to measure.

From Output to Value Added

A firm's **total revenue** is $P_i Y_i$. Its costs include:

- **Intermediate inputs** (X_i): pharmaceuticals, medical supplies, purchased services, utilities
- **Factor payments**: wages to labor ($w_i L_i$) and returns to capital ($r_i K_i$)

Value added is the firm's contribution net of intermediate inputs:

$$\text{VA}_i = P_i Y_i - P_X X_i$$

Value Added = Factor Payments

Under competitive factor markets, value added equals the total payments to primary factors:

$$\text{VA}_i = \underbrace{w_i L_i}_{\text{labor payments}} + \underbrace{r_i K_i}_{\text{capital payments}}$$

Factor Shares in Theory

The **labor share** and **capital share** measure the fraction of value added paid to each factor:

$$\alpha_L^i = \frac{w_i L_i}{\text{VA}_i}, \qquad \alpha_K^i = \frac{r_i K_i}{\text{VA}_i}, \qquad \alpha_L^i + \alpha_K^i = 1$$

Under Cobb-Douglas ($Y = AK^\alpha L^{1-\alpha}$) with competitive markets, these shares are **constant**:

$$\alpha_K = \alpha, \qquad \alpha_L = 1 - \alpha$$

Empirical Questions

- Are healthcare factor shares constant over time, or are they changing?
- Does healthcare differ from other industries?
- How sensitive are the estimates to the treatment of mixed income?

Value Added in the National Accounts

In the national accounts, value added maps to three measured components:

$$\mathrm{VA}_i = \underbrace{\mathrm{CE}_i}_{\text{labor}} + \underbrace{\mathrm{GOS}_i}_{\text{capital}} + \underbrace{\mathrm{TOPI}_i}_{\text{taxes}}$$

Components

CE Compensation of Employees — wages, salaries, and supplements (pensions, health insurance, social insurance)

GOS Gross Operating Surplus — returns to capital (profits, depreciation, net interest)

TOPI Taxes on Production and Imports — less subsidies

Source: BEA GDP-by-Industry accounts (NAICS basis, 1997–2024).

Raw Factor Shares

Dividing each component by value added yields factor shares:

$$\alpha_L^i = \frac{\text{CE}_i}{\text{VA}_i} \qquad \text{(labor share)}$$

$$\alpha_K^i = \frac{\text{GOS}_i}{\text{VA}_i} \qquad \text{(capital share)}$$

$$\alpha_T^i = \frac{\text{TOPI}_i}{\text{VA}_i} \qquad \text{(tax share)}$$

Accounting Identity

$$\alpha_L^i + \alpha_K^i + \alpha_T^i = 1 \qquad \forall\ i$$

This identity holds exactly in every industry-year cell by construction.

Data Sources

Primary: BEA GDP-by-Industry

- Value added components (CE, GOS, TOPI) by NAICS industry
- 15 industries including healthcare subsectors (621, 622, 623)
- Annual data, 1997–2024 (420 industry-year observations)

Supplementary

- **NIPA Tables:** Proprietors' income (Table 6.12), self-employment counts (Table 6.7), FTE employees (Table 6.5)
- **BLS QCEW:** Employment levels and wages by industry and ownership (validation)

The Mixed Income Problem

The BEA decomposition assigns proprietors' income (M_i) to GOS:

$$\text{VA}_i = \text{CE}_i + \text{GOS}_i + \text{TOPI}_i \quad \text{where GOS includes } M_i$$

But proprietors' income is **mixed** — it contains returns to both labor *and* capital:

$$M_i = \underbrace{M_i^L}_{\substack{\text{labor}\\ \text{component}}} + \underbrace{M_i^K}_{\substack{\text{capital}\\ \text{component}}}$$

Implication

Raw $\alpha_L = \text{CE}/\text{VA}$ **understates** the true labor share, especially in industries with significant self-employment (e.g., physician practices).

Following Gollin2002).

Gollin Adjustment: Proportional Allocation

Allocate proprietors' income between labor and capital in the same ratio as the corporate sector (Gollin2002):

Step 1. Corporate-sector labor share: $$s_i = \frac{\text{CE}_i}{\text{VA}_i - M_i - \text{TOPI}_i}$$

Step 2. Impute labor component of proprietors' income: $$\text{CE}_i^* = \text{CE}_i + M_i \times s_i$$

Step 3. Adjusted labor share: $$\alpha_L^{\text{adj},i} = \frac{\text{CE}_i^*}{\text{VA}_i}$$

Intuition: If the corporate sector allocates 70% of non-proprietor income to labor, assume proprietors do the same.

Gollin Adjustments: Bounding Methods

Method 2: All-Labor Upper Bound. Treat *all* proprietors' income as labor returns:

$$\alpha_L^{\text{upper},i} = \frac{\text{CE}_i + M_i}{\text{VA}_i}$$

This provides an upper bound on the labor share.

Method 3: Imputed Wage. Assume proprietors earn average employee compensation:

$$\alpha_L^{\text{imputed},i} = \frac{\text{CE}_i + N_i^{\text{self}} \times \bar{w}_i}{\text{VA}_i}$$

where $\bar{w}_i = \text{CE}_i / \text{FTE}_i$ is compensation per full-time equivalent employee.

Mixed Income: Quantitative Impact

Industry	Raw (CE/VA)	Gollin Prop.	All-Labor
Healthcare (62)	0.813	**0.902**	0.909
Construction (23)	0.639	0.834	0.869
Professional svcs (54)	0.669	0.811	0.840
Retail (44–45)	0.545	0.593	0.611
Manufacturing (31–33)	0.515	0.520	0.527
Information (51)	0.382	0.389	0.404
Total economy	0.539	—	—

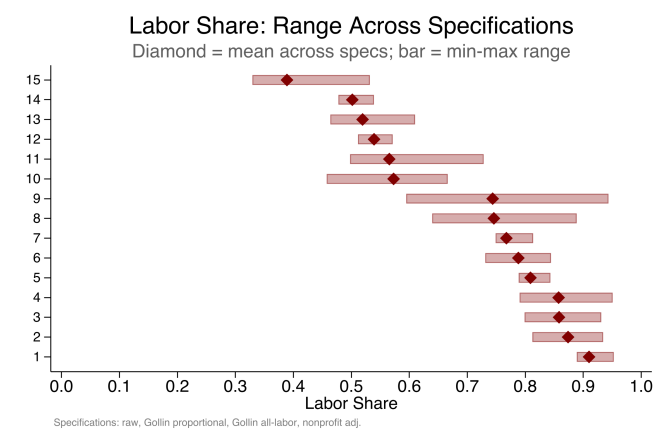
Table 1: *

Mean labor shares, 1997–2024.

Key Insight

The Gollin adjustment raises healthcare's labor share from 0.81 to 0.90. The effect is largest where self-employment is significant; manufacturing is largely unaffected.

Sensitivity to Mixed Income Treatment



Diamond = mean across specifications; bar = min–max range.

Sensitivity: Nonprofit Capital Returns

For hospitals (NAICS 622), approximately 75% are nonprofit institutions. GOS in nonprofits reflects cost recovery and retained surplus, not economic profit. We adjust:

$$\alpha_K^{\text{adj},i} = \frac{\text{GOS}_i}{\text{VA}_i} \times (1 - \pi_{\text{np}} \times \varphi)$$

where $\pi_{\text{np}} = 0.75$ (nonprofit share) and $\varphi = 0.50$ (cost recovery fraction).

Hospital Example

- Raw capital share: $\alpha_K = 0.165$
- Nonprofit-adjusted: $\alpha_K^{\text{adj}} = 0.165 \times (1 - 0.75 \times 0.50) = 0.103$
- Implied labor share rises from 0.835 to 0.897 (abstracting from taxes)

Sensitivity: Gross vs. Net of Depreciation

The baseline uses *gross* operating surplus. An alternative subtracts depreciation:

$$\text{NOS}_i = \text{GOS}_i - \text{CFC}_i \qquad \text{(Net Operating Surplus)}$$

$$\text{NVA}_i = \text{VA}_i - \text{CFC}_i \qquad \text{(Net Value Added)}$$

$$\alpha_K^{\text{net},i} = \frac{\text{NOS}_i}{\text{NVA}_i} \qquad \text{(Net capital share)}$$

Interpretation

- CFC = Consumption of Fixed Capital (depreciation of equipment, structures, IP)
- Net specification: returns to capital *after* replacing worn-out capital
- Relevant for capital-intensive healthcare subsectors (hospitals, imaging centers)

Requires CFC data from NIPA Fixed Asset supplements (not yet available at industry detail).

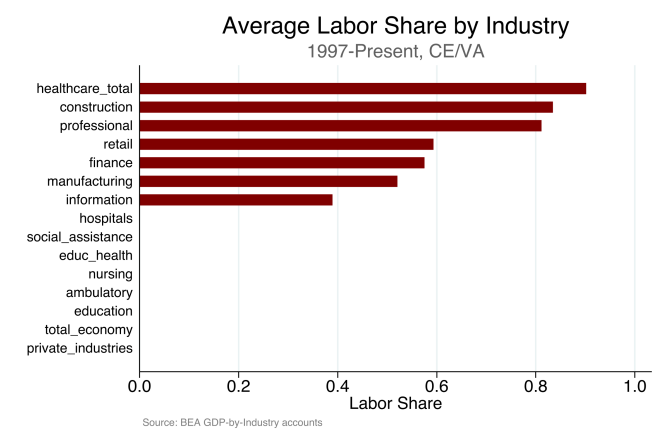
Robustness Across Specifications

Industry	Raw (CE/VA)	Gollin Proportional	All-Labor Bound	Nonprofit Adjusted
Healthcare (62)	0.813	0.902	0.909	0.813
Ambulatory (621)	0.767	—	—	0.767
Hospitals (622)	0.835	—	—	0.913
Nursing (623)	0.910	—	—	0.910
Manufacturing (31–33)	0.515	0.520	0.527	0.515
Finance (52)	0.548	0.575	0.592	0.548
Information (51)	0.382	0.389	0.404	0.382
Professional (54)	0.669	0.811	0.840	0.669
Total economy	0.539	—	—	0.539

Table 2: *

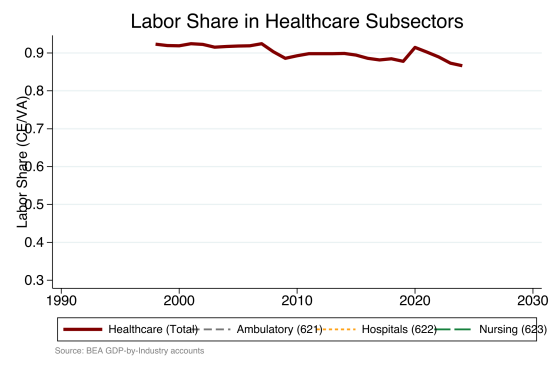
Mean labor shares, 1997–2024. "—" = data unavailable at this NAICS detail.

Labor Shares Across Industries



Healthcare subsectors cluster at the top — among the most labor-intensive industries in the U.S. economy.

Healthcare Subsectors Over Time



- Nursing & residential care (623): consistently highest (∼0.91)
- Hospitals (622): middle (∼0.84), capital in buildings and equipment
- Ambulatory care (621): lowest within healthcare (∼0.77)

Value Added vs. Gross Output Across Industries

Value added is a fraction of gross output; the rest is intermediate inputs ($GO = VA + II$):

Industry	VA ($B)	GO ($B)	VA/GO	VA/GDP
Healthcare (62)	2,044	3,294	**0.620**	7.3%
Ambulatory (621)	1,005	1,483	0.678	3.6%
Hospitals (622)	643	1,161	0.554	2.3%
Nursing (623)	197	317	0.621	0.7%
Manufacturing (31–33)	2,817	7,143	0.394	10.1%
Professional svcs (54)	2,221	3,295	0.674	8.0%
Information (51)	1,491	2,537	0.588	5.4%

BEA GDP-by-Industry, 2023. Healthcare retains 62% of GO as VA—higher than manufacturing (39%) but lower than professional services (67%). Hospitals have the lowest VA/GO (0.55), reflecting large intermediate input purchases.

Healthcare Spending $\neq$ Healthcare Value Added

CMS National Health Expenditures ($\sim$18% of GDP) spans *multiple* BEA industries:

BEA Industry	VA ($B)	VA/GDP
Health care & social assistance (62)	2,044	7.3%
Chemical products, incl. pharma (325)	515	1.9%
Misc. mfg, incl. medical devices (339)	109	0.4%
Insurance carriers, incl. health ins. (524)	750	2.7%
Wholesale trade, incl. drug distribution (42)	1,653	5.9%

Table 3: *

2023. Only a fraction of each affiliated industry is healthcare-related.

- Healthcare **value added** (NAICS 62) = 7.3% of GDP—our measurement base
- Remaining $\sim$11% of NHE reflects intermediate inputs *and* output of affiliated industries (pharma, devices, health insurance, drug wholesaling)

Reconciling Labor Share, Employment, and Wages

The Apparent Puzzle

Healthcare has a very high labor share (0.81), yet produces 7.3% of GDP with ∼12% of workers.

	Healthcare	Economy
Share of GDP (value added)	7.3%	100%
Share of employment	∼12%	100%
VA per worker (relative)	∼0.6×	1×
Labor share (α_L)	0.81	0.54
Avg. compensation (relative)	∼1.15×	1×

Resolution

High labor share means most VA *accrues to workers*, not that there are many. Healthcare has **below**-average VA per worker but **above**-average wages:

$$\frac{w_{\text{hc}}}{w_{\text{avg}}} = \frac{\alpha_L^{\text{hc}}}{\alpha_L^{\text{avg}}} \times \frac{\text{VA/worker}_{\text{hc}}}{\text{VA/worker}_{\text{avg}}} \approx \frac{0.81}{0.54} \times 0.6 \approx 1.15$$

Methodology: Summary

Main Findings

1. **Healthcare is highly labor-intensive:** raw labor share of 0.81, rising to 0.90 after the Gollin adjustment for proprietors' income
2. **Heterogeneity within healthcare:** nursing facilities (0.91) vs. ambulatory care (0.77) — a 14 percentage point gap
3. **Results are robust:** labor share ranges from 0.81 to 0.91 across all specifications (raw, Gollin, nonprofit-adjusted)
4. **Healthcare stands apart:** far above the total economy (0.54) and capital-intensive sectors like information (0.38) and manufacturing (0.52)

Thank you